DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

END-SEM EXAMINATION- Summer 2019

Course: B. Tech in Electronics and Telecommunication Engineering **Sem: III**

Subject Name: Network Analysis **Subject Code: BETX304**

Max Marks: 60 **Date:31/05/2019** **Duration: 3 Hr.**

Instructions to the Students:

1. *Solve **ANY FIVE** questions out of the following.*
2. *The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve the following. **CO01**

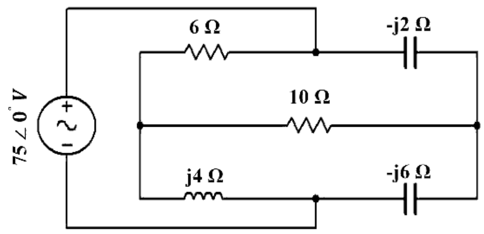
A) Explain the terms **[06]**

(a) Passive network and active network

(b) Linear Element and Nonlinear element

(c) Lumped Element and distributed element

B) With the help of mesh analysis find current through 6 Ω capacitor show in figure. **[06]**



Q.2 Solve. **CO02**

A) Derive expression for frequencies at which voltage across L and C are maximum in series resonant circuit. **[08]**

B) A series RLC circuit with resistance of 10 Ω inductance of 0.2 H and capacitance of 100 μF is applied by voltage source of 50 V. Find resonant frequency. **[04]**

Q. 3 Solve any one the following.

A) Design a composite high pass filter to operate into load of 600 Ω and have cutoff frequency of 1.6 kHz. The filter should on constant k section, one m derived section with frequency of infinite attenuation is 1.3kHz and a suitable termination half section. **[12]**

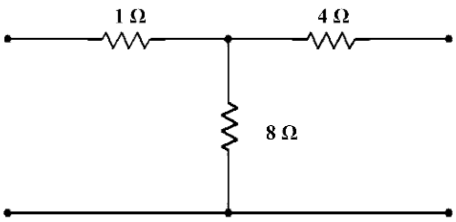
B) I) Define characteristic impedance. Derive and expression for characteristic impedance of symmetrical T network in terms of open and short circuit impedances. **[06]**

II) Design a prototype high pass filter section if design impedance R_0 = 300 Ω and cut off frequency (a) f_c = 100 Hz, (b) f_c = 200 Hz. **[06]**

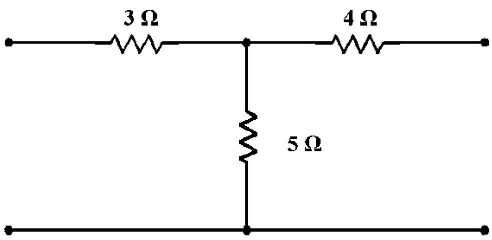
Q.4 Solve any two of the following. **CO01**

A) Define: (a) forward transfer admittance, (b) forward transfer impedance, (c) voltage transfer ratio and (d) current transfer ratio. **[06]**

B) For the network shown below determine ABCD (transmission) parameter. **[06]**

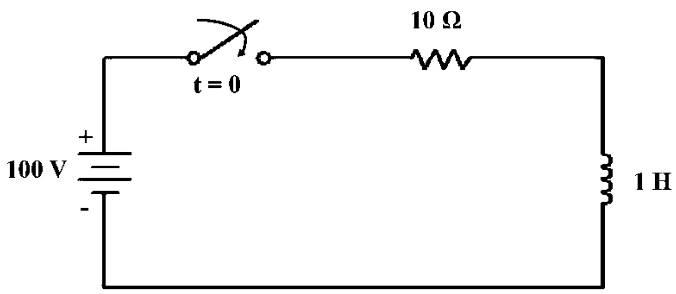


C) For the network shown below determine Y (short circuit admittance) parameter. Also verify condition for reciprocity and symmetry for the same. **[06]**



Q. 5 Solve the following. **CO01,CO03**

A) The switch is closed at t = 0 find value of $i, \frac{di}{dt}, \frac{d^2i}{dt^2}$ at $t = 0^+$. Assume initial current in the inductor to be zero. **[06]**



B) Explain behavior of basic elements (R, L and C) in Laplace domain. **[06]**

Q. 6 Solve any two of the following **CO03,CO04**

A) Explain in brief: (a) types of transmission lines and (b) parameters of transmission line. **[06]**

B) What is distortion less line? Derive the condition for a distortion less transmission line. **[06]**

C) a transmission line has following parameters at 1 kHz frequency $R = 4\ \Omega/km$, $L = 1\ mH/km$, $G = 0.5 \times 10^{-6}$ mho/km and $C = 0.001\ \mu F/km$. Calculate Z_o, attenuation constant, phase constant, wavelength, phase velocity. **[06]**

***** End *****